

Trainings ▾

Fully Formulated Coolant/Antifreeze

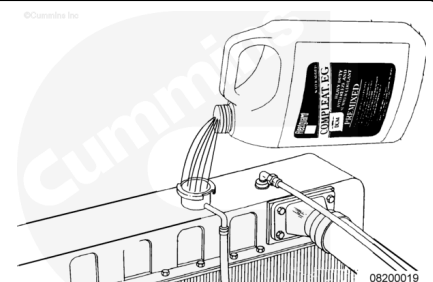
For new vehicles, check with the original equipment manufacturer (OEM) on the type of coolant (ethylene glycol, propylene glycol, organic acid technology, glycerin, etc.) used for the first fill. This will assist in understanding how to maintain the coolant properly.

Cummins Inc. recommends using either a 50/50 mixture of good quality water and fully formulated antifreeze or fully formulated coolant when filling the cooling system. The fully formulated antifreeze or coolant **must** meet CES14603 specifications.

Most coolants which meet ASTM D6210 also meet CES14603.

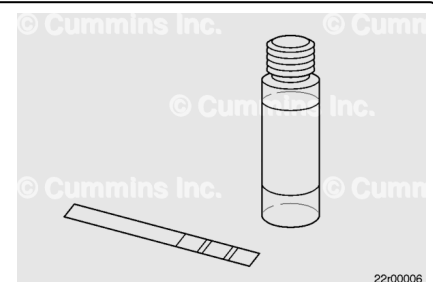
However, some organic acid technology coolants such as Shell™ Rotell ELC, Chevron™ Texaco™ Delo ELC, and their private label counterparts meet ASTM D6210, but do **not** meet the elastomer compatibility test of CES14603. These coolants are acceptable for use, assuming the OEM added silicate at initial fill. Refer to Bulletin 3666132

(/qs3/pubsys2/xml/en/bulletin/3666132.html), Cummins® Coolant Requirements and Maintenance, Section 3, Extended Service Interval, for more details.



Good quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems and excessive levels of chlorides and sulfates cause cooling system corrosion.

Use Fleetguard® Water-Chek™ test strips, CC2609, to check the water quality. Instructions are included with the test kit.



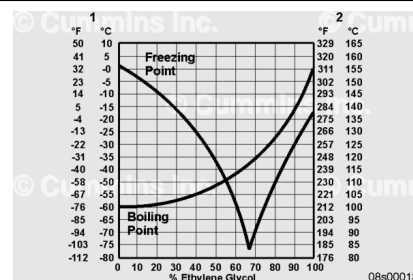
Water Quality	
Calcium Magnesium (hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	Maximum 40 ppm as (Cl)
Sulfate	Maximum 100 ppm as (SO ₄)

Cummins Inc. recommends Cummins Filtration™ antifreeze coolants including Compleat ES™ containing DCA4 Plus, Fleetcool™ EX containing DCA2 Plus, and ES Optimax™ organic acid technology, which meet the requirements of Cummins® Engineering Standard 14603.



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Fully formulated antifreeze **must** be mixed with good quality water at a 50/50 ratio (40 to 60 percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-33°F] freezing point and a 108°C [226°F] boiling point, which is adequate for locations in North America. The actual lowest freezing point of ethylene glycol antifreeze is at 68 percent. Using higher concentrations of antifreeze will raise the freezing point of the solution and increase the possibility of a silica gel problem.



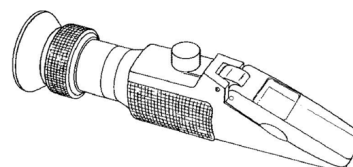
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Legend

1. Freezing Point Temperature Scale
2. Boiling Point Temperature Scale

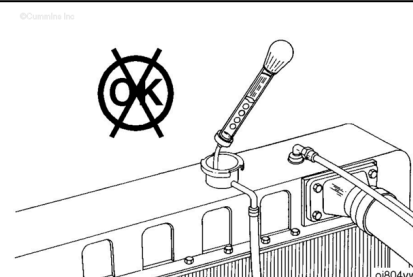
A refractometer is the preferred tool for measuring the freezing point of ethylene glycol and propylene glycol coolants. Use Fleetguard® refractometer, Part Number CC2800 or CC2806. For glycerin coolants, use Part Number CC36049.

Note : Fleetguard® coolant test strips, Part Number CC2602, can also be used to check the freezing point.



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Do **not** use a floating ball hydrometer. The use of a floating ball hydrometer can give incorrect readings.

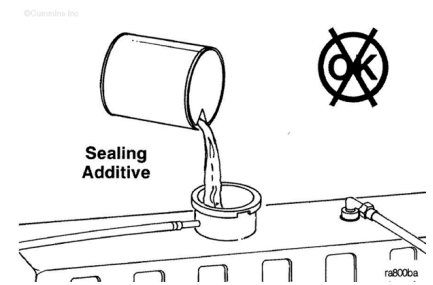


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Cooling System Sealing Additives

Do **not** use sealing additives in the cooling system. The use of sealing additives will:

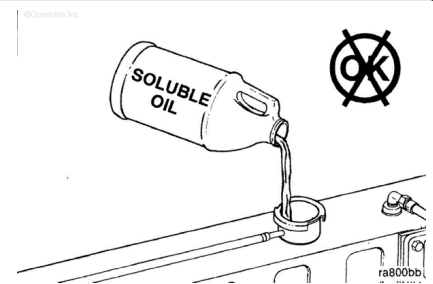
- Build up in coolant low-flow areas
- Clog coolant filters, if equipped
- Plug radiators and oil coolers
- Possibly damage the water pump seal.



Cooling System Soluble Oils

Do **not** use soluble oils in the cooling system. The use of soluble oils will:

- Allow cylinder liner pitting
- Corrode brass and copper
- Damage heat transfer surfaces
- Damage seals and hoses.

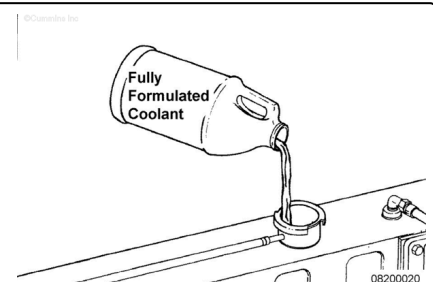


Supplemental Coolant Additive (SCA)

⚠ CAUTION ⚠

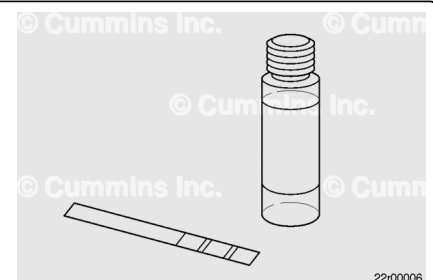
Improper coolant, concentration of coolant additives, and maintenance will likely result in liner pitting and engine failure.

Fully formulated products contain SCAs and are required to protect the cooling system from scale and fouling, solder corrosion, and general corrosion.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Measuring the chemical protection in the engine's cooling system is essential for protection against liner pitting, corrosion, and coolant dilution.

Note : Some organic acid technology coolants use high levels of organic acids for liner pitting protection and do not use nitrite and/or molybdate. Therefore, these coolants do not have SCA numbers. See the coolant manufacturer for proper maintenance instructions.

Testing is especially recommended if the operator is **not** sure of the cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

If the SCA concentration is below 1.2 units per gallon, the supplemental coolant additives need to be replenished.

Check the SCA concentration level at least every 6 months, and anytime the coolant condition is unknown or corrosion is apparent within the cooling system.

Use Fleetguard® coolant test strips, Part Number CC2602, to check the concentration level. Instructions are included with the test kit.

Call the following numbers to get answers to any questions about cooling system maintenance:

Cummins Inc.	Fleetguard®
1-800-DIESELS	1-800-22FILTER
1-800-343-7357	1-800-223-4583

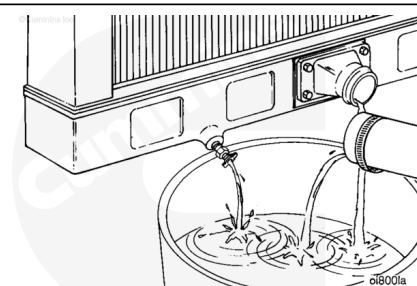
Coolant Replacement Requirements

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Replace the coolant **only** if the replacement limits are exceeded.

Fleetguard® Quik-Chek™ test strips, Part Number CC2718, will detect contamination levels that indicate replacement of the coolant is required.



Fleetguard® also offers laboratory coolant testing with the use of Monitor Coolant Testing kits, Part Numbers CC2700 and CC2706.

Note : Dispose of used coolant or antifreeze in accordance with federal, state, and local laws and regulations.

Last Modified: 30-Jul-2015
